

5.1 Parametric Equations

Not all curves pass the vertical line test. It is impossible to describe such curves in the form $y = f(x)$. The idea in this section is to express such curves in the xy -plane with parametric equations.

Terminology: Suppose x and y are both given as functions of a third variable t (called a parameter). In terms of equations we write:

$$x = f(t) \quad y = g(t)$$

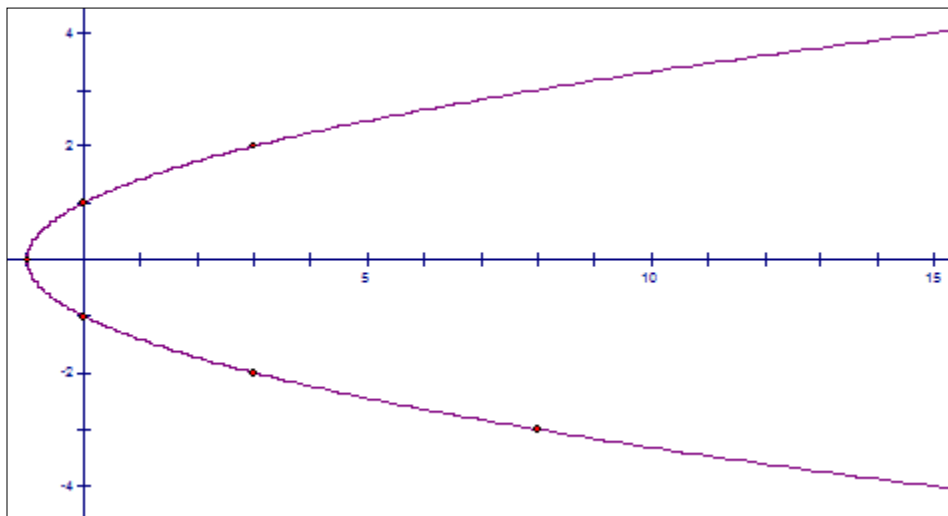
and we call these equations parametric equations. Each value of t determines a point (x, y) which we can plot in the plane. As t varies we can trace out a curve determined by the parametric equations which we will call the parametric curve.

Example 1: (a) Sketch the curve given by the parametric equations: $x = t^2 - 2t$ $y = t - 1$.

We begin by creating a table to identify some points to graph:

t	x	y
-2	8	-3
-1	3	-2
0	0	-1
1	-1	0
2	0	1
3	3	2

Now, the graph is:



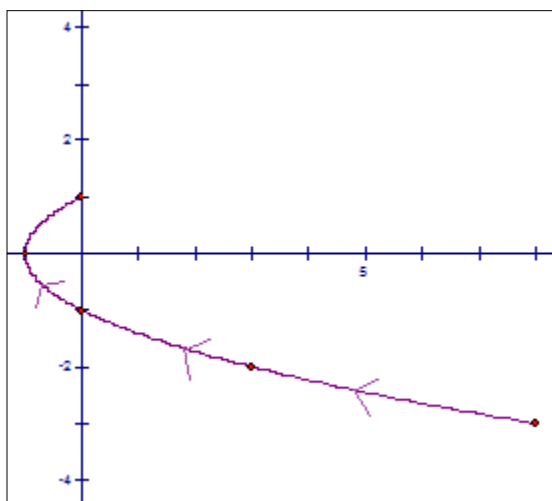
(b) Describe this curve with a single equation which does not have the parameter t in it.

We note that $t = y + 1$. So, if we perform a substitution we find that an equation which describes the curve is: $x = (y + 1)^2 - 2(y + 1)$.

Example 2: One nice application of parametric equations is that they can be used to model the movement of a particle that traverses some curve in the plane (in some applications we view t as the time variable). Draw the path traversed by a particle whose movement is modeled by the parametric equations:

$$x = t^2 - 2t \quad y = t - 1 \quad \text{where } -2 \leq t \leq 2.$$

For this question the path is part of the graph of the previous example with an orientation:



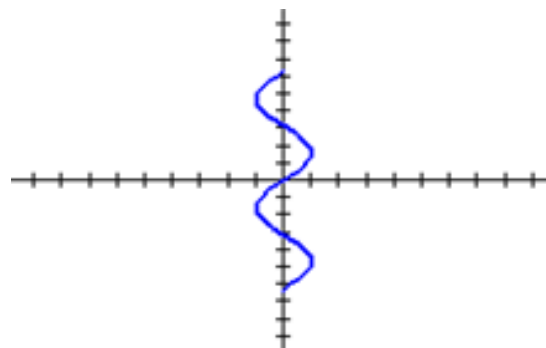
Note: In the above example the point $(8, -3)$ (where the particle starts) is called the initial point, and the point $(0, 1)$ (where the particle ends) is called the terminal point.

Note: Many graphs that cannot be graphed in xy -coordinates can be expressed as parametric equations.

Example 3: Graph $x = \sin(y)$ $-2\pi \leq y \leq 2\pi$ parametrically.

We first need to convert this equation to a parametrically defined equation.

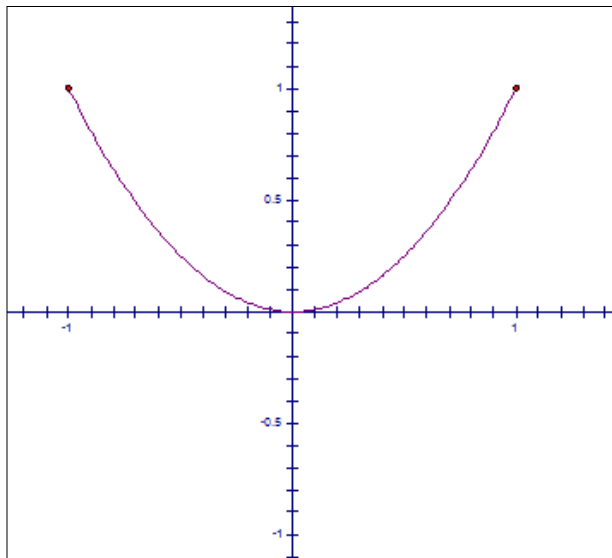
If we let $y = t$ then we obtain: $\begin{cases} x = \sin(t) \\ y = t \end{cases}$ for $-2\pi \leq t \leq 2\pi$



Example 4 (you try): (unrelated parts)

- (a) Sketch the curve with parametric equations: $x = \cos(t)$ $y = \cos^2(t)$

We note that these equations imply $y = x^2$, but since the range values of cosine are in $[-1, 1]$, we only get part of this parabola. The answer is:



- (b) Find parametric equations for the curve determined by the equation: $x = y^4 - 3y^2$

If we let $y = t$, we obtain the parametric equations: $x = t^4 - 3t^2$ $y = t$.

- (c) Find parametric equations which describe the movement of a particle which traverses a circle of radius 4 exactly one time in the counterclockwise direction.

One possibility is: $x = 4\cos(t)$ $y = 4\sin(t)$ where $0 \leq t \leq 2\pi$.

Calculus with Parametric Curves

Motivation: Suppose f and g are differentiable functions and we want to find the tangent line at a point on the curve $x = f(t)$, $y = g(t)$. If we suppose that y is a differentiable function of x . Then, we know by the chain rule:

$$\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}.$$

Now, we can solve for $\frac{dy}{dx}$ provided that $\frac{dx}{dt} \neq 0$.

Note: The subtlety here is that in general y may not be a function of x . We said early that one of the main reasons we use parametric equations is due to the fact that y is not a function of x . In more advanced mathematics (namely real analysis) one can prove that, when $\frac{dx}{dt} \neq 0$, then “ y is locally a function of x ” (the definition of locally is beyond the scope of the course). This is great news because it leads to the following theorem which gives us a rule for differentiation.

Theorem: If f and g are differentiable functions with $x = f(t)$, $y = g(t)$ and $\frac{dx}{dt} \neq 0$, then

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

Note: The proof of this theorem is beyond the scope of the course.

Note: The parametric curve given by $x = f(t)$, $y = g(t)$ has a horizontal tangent when $\frac{dy}{dt} = 0$ provided $\frac{dx}{dt} \neq 0$. The parametric curve given by $x = f(t)$, $y = g(t)$ has a vertical tangent when $\frac{dx}{dt} = 0$ provided $\frac{dy}{dt} \neq 0$. When both $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are 0 a careful analysis is needed (see 7.4.1).

Note: $\frac{d^2y}{dx^2} \neq \frac{\frac{d^2y}{dt^2}}{\frac{d^2x}{dt^2}}$ rather $\frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{dy}{dx} \right) = \frac{\frac{d}{dt} \left(\frac{dy}{dx} \right)}{\frac{dx}{dt}}$

Example 5: Suppose that a curve C is defined by the parametric equations $x = t^2$, $y = t^3 - 9t$.

(a) Find the equation of both tangent lines to C at $(9, 0)$.

To find the value(s) of t which correspond to the point $(9, 0)$ we solve $9 = t^2 \Leftrightarrow t = 3, -3$. We notice that the t values of both 3 and -3 correspond to the point $(9, 0)$ on C .

By the above theorem we know that for $t \neq 0$, $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{3t^2 - 9}{2t} = \frac{3}{2} \left(t - \frac{3}{t} \right)$. This means there is a tangent

line at $(9, 0)$ with slope $\frac{3}{2} \left(3 - \frac{3}{3} \right) = 3$ and a tangent line at $(9, 0)$ with slope $\frac{3}{2} \left(-3 - \frac{3}{-3} \right) = -3$.

Now, we use point slope form of a line to write the equations of both tangent lines at $(9, 0)$. In particular the answer is: $y = 3(x - 9)$ and $y = -3(x - 9)$.

(b) Find the point(s) on C where a tangent is horizontal or vertical.

A tangent will be horizontal for the point(s) corresponding to values of t which satisfy: $t \neq 0$ and

$\frac{dy}{dt} = 0 \Leftrightarrow 3t^2 - 9 = 0 \Leftrightarrow t^2 = 3 \Leftrightarrow t = \pm\sqrt{3}$. Now, we plug these values of t into the parametric equations and

we find that there is a horizontal tangent at the points $(3, -6\sqrt{3})$ and $(3, 6\sqrt{3})$.

A tangent will be vertical for the point(s) corresponding to values of t which satisfy: $t \neq \pm\sqrt{3}$ and

$\frac{dx}{dt} = 0 \Leftrightarrow 2t = 0 \Leftrightarrow t = 0$. Now, we plug this value of t into the parametric equations and we find that there is

a vertical tangent at the point $(0, 0)$.

(c) For what values of t is the curve concave upward/downward.

First we calculate: $\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt}\left(\frac{3}{2}\left(t - \frac{3}{t}\right)\right)}{2t} = \frac{\frac{3}{2}\left(1 + \frac{3}{t^2}\right)}{2t} \cdot \frac{2t^2}{2t^2} = \frac{3(t^2 + 3)}{4t^3}$ provided $t \neq 0$. Now, we

notice that $\frac{d^2y}{dx^2}$ is positive for $t > 0$ and $\frac{d^2y}{dx^2}$ is negative for $t < 0$. This means that C is concave upward for $t > 0$, and C is concave downward for $t < 0$.

Note: One can check the reasonableness of the answers by graphing the curve on the calculator.

Example 6 (you try):

For which values of t is the curve given by $x = t^2 + 1$, $y = e^t - 1$ concave upward/downward.

By the theorem we know that for $t \neq 0$ we have $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{e^t}{2t}$. Then, for $t \neq 0$ we have:

$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}} = \frac{\frac{d}{dt}\left(\frac{e^t}{2t}\right)}{2t} = \frac{\left(\frac{2te^t - 2e^t}{4t^2}\right)}{2t} = \frac{te^t - e^t}{4t^3} = \frac{e^t(t-1)}{4t^3}$. Now, we want to know all the values of t

($t \neq 0$) such that $\frac{e^t(t-1)}{4t^3} < 0$. We note that the numerator of the expression on the left is positive for $t > 1$ and negative for $t < 1$. Also the denominator of the expression on the left is positive for $t > 0$ and negative for $t < 0$.

The curve is concave downward for $0 < t < 1$.

The curve is concave upward for $t > 1$ or $t < 0$.

Using the ideas from chapter 2 & 3 we can find ways to compute areas beneath/between parametric curves and derive formulas for arc length and surface area in the setting of parametric equations. We will only discuss arc length. I will include extra (optional) notes on the other topics mentioned above and other details are in the textbook.

Theorem: If a curve C is described by the parametric equations $x = f(t)$ $y = g(t)$, $\alpha \leq t \leq \beta$, where f' and g' are continuous on $[\alpha, \beta]$ and C is traversed exactly once as t increases from α to β , then the arc length of C is

$$L = \int_{\alpha}^{\beta} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt .$$

Proof: Shown in the textbook.

Example 7: Find the exact length of the curve given by $x = e^t + e^{-t}$ $y = 5 - 2t$ where $0 \leq t \leq 3$.

We note that $\frac{dx}{dt} = e^t - e^{-t}$ and $\frac{dy}{dt} = -2$. We note that both of these derivatives are continuous on $[0, 3]$, and since the function for y is a decreasing function we know that the curve is only traversed once as t increases from 0 to 3. Thus, the above theorem applies and the length is:

$$\begin{aligned} L &= \int_0^3 \sqrt{(e^t - e^{-t})^2 + (-2)^2} dt = \int_0^3 \sqrt{e^{2t} - 1 - 1 + e^{-2t} + 4} dt = \int_0^3 \sqrt{e^{2t} + 2 + e^{-2t}} dt \\ &= \int_0^3 \sqrt{(e^t + e^{-t})^2} dt = \int_0^3 (e^t + e^{-t}) dt = (e^t - e^{-t}) \Big|_0^3 = e^3 - e^{-3} . \end{aligned}$$

Example 8 (you try): Set up an integral that represents the length of the curve given by $x = t^2 - t$ $y = t^4$ where $1 \leq t \leq 4$. Then, use your calculator to find the approximate length to four decimal places.

Solution: We note that $\frac{dx}{dt} = 2t - 1$ and $\frac{dy}{dt} = 4t^3$. We note that both of these derivatives are continuous on $[1, 4]$, and since the function for y is increasing on $[1, 4]$, we know that the curve is only traversed once as t increases from 1 to 4. Thus, the above theorem applies and the length is:

$$L = \int_1^4 \sqrt{(2t-1)^2 + (4t^3)^2} dt \approx 255.3756 .$$

OPTIONAL MATERIAL: NOT COVERED ON HOMEWORK OR EXAMS.

Now we will explore integration and areas under curves given by parametric equations.

Definition: Suppose we have a circle of radius r with center $(0, r)$. Fix point P as the point on the circle which coincides with the origin. The curve that P traverses as the circle rolls along the x -axis is known as a cycloid.

Fact: The parametric equations of the cycloid are: $x = r(\theta - \sin(\theta))$ $y = r(1 - \cos(\theta))$

Note: Details on cycloids can be found on in the textbook.

Example 1: Use your calculator to graph the cycloid given by: $x = \theta - \sin(\theta)$ $y = 1 - \cos(\theta)$.

The key thing to remember here is that you may have to go to window and adjust the T minimum and T maximum in order to get a full picture. Also if your T step is too far apart your picture may not be accurate.

Example 2: Find the area under one arch of the cycloid given by $x = r(\theta - \sin(\theta))$ $y = r(1 - \cos(\theta))$.

From the graph of a cycloid, we know that one arch of the cycloid is traced as θ varies from 0 to 2π . We also know that this arch of the cycloid starts at the origin and ends at the point $(2\pi r, 0)$ in the xy -plane.

From Calculus 1 we know that the desired area is given by the definite integral: $\int_0^{2\pi r} y \, dx$. This integral is tough to evaluate since y is not written in terms of x . So, we rewrite the integral in terms of θ .

To do this, we note that $dx = r(1 - \cos(\theta))d\theta$, and by what we have already mentioned, the new limits of integration will be 0 and 2π (since one arch of the cycloid is traced as θ varies from 0 to 2π). Now, we have:

$$\begin{aligned}
\int_0^{2\pi r} y \, dx &= \int_0^{2\pi} r(1 - \cos(\theta)) r(1 - \cos(\theta)) \, d\theta = r^2 \int_0^{2\pi} (1 - \cos(\theta))^2 \, d\theta \\
&= r^2 \int_0^{2\pi} (1 - 2\cos(\theta) + \cos^2(\theta)) \, d\theta = r^2 \int_0^{2\pi} \left(1 - 2\cos(\theta) + \frac{1}{2}(1 + \cos(2\theta))\right) \, d\theta \\
&= r^2 \int_0^{2\pi} \left(\frac{3}{2} - 2\cos(\theta) + \frac{1}{2}\cos(2\theta)\right) \, d\theta = r^2 \left[\frac{3}{2}\theta - 2\sin(\theta) + \frac{1}{4}\sin(2\theta)\right]_0^{2\pi} \\
&= r^2 \left[3\pi - 2\sin(2\pi) + \frac{1}{4}\sin(4\pi) + 2\sin(0) - \frac{1}{4}\sin(0)\right] = 3\pi r^2.
\end{aligned}$$

Example 3 (you try): Find the area enclosed by the curve given by $x = t^2 - 2t$ $y = \sqrt{t}$ and the y -axis.

We see that portion of the curve that we are interested in is traced as t varies from 0 to 2. We see that the portion of the curve traced as t varies from 0 to 2 starts at the origin and ends at $(0, \sqrt{2})$. The (signed) area that

we are interested in is given by the integral: $\int_0^{\sqrt{2}} x \, dy$. We note that $dy = \frac{1}{2\sqrt{t}} \, dt$ (assuming that $t > 0$). Thus, the

$$\int_0^2 (t^2 - 2t) \frac{1}{2\sqrt{t}} \, dt = \lim_{n \rightarrow 0^+} \left(\int_n^2 (t^2 - 2t) \frac{1}{2\sqrt{t}} \, dt \right) = \lim_{n \rightarrow 0^+} \left(\frac{1}{2} \int_n^2 \left(t^{\frac{3}{2}} - 2t^{\frac{1}{2}} \right) \, dt \right)$$

integral becomes
$$\begin{aligned}
\lim_{n \rightarrow 0^+} \left(\frac{1}{2} \left[\frac{2}{5} t^{\frac{5}{2}} - \frac{4}{3} t^{\frac{3}{2}} \right]_n^2 \right) &= \lim_{n \rightarrow 0^+} \left(\frac{1}{2} \left[\frac{2}{5} 4\sqrt{2} - \frac{4}{3} 2\sqrt{2} - \left(\frac{2}{5} n^{\frac{5}{2}} - \frac{4}{3} n^{\frac{3}{2}} \right) \right] \right) \\
&= \frac{4\sqrt{2}}{5} - \frac{4\sqrt{2}}{3} = \frac{-8\sqrt{2}}{15}.
\end{aligned}$$

Assuming that we want a positive area (and not a signed area) the answer is: $\frac{8\sqrt{2}}{15}$.

Theorem: If a curve C is described by the parametric equations $x = f(t)$ $y = g(t)$ where $\alpha \leq t \leq \beta$ is rotated about the x -axis, where f' and g' are continuous on $[\alpha, \beta]$ and $g(t) \geq 0$, then area of the resulting surface is

$$S = \int_{\alpha}^{\beta} 2\pi y \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \, dt.$$

Note: In this theorem we have $ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \, dt$

Example 4: Find the exact surface area of the surface obtained by rotating the curve given by $x = e^t + e^{-t}$ $y = 5 - 2t$ where $0 \leq t \leq 2$ about the x -axis.

From a previous example we know that $\frac{dx}{dt}$ and $\frac{dy}{dt}$ are continuous on $[0, 2]$. We also know that the function describing y is nonnegative on $[0, 2]$. Thus, the above theorem applies and the surface area is:

$$\begin{aligned}
 L &= 2\pi \int_0^2 (5-2t) \sqrt{(e^t - e^{-t})^2 + (-2)^2} dt = 2\pi \int_0^2 (5-2t) \sqrt{(e^t + e^{-t})^2} dt \\
 &= 2\pi \int_0^2 (5-2t)(e^t + e^{-t}) dt = 10\pi \int_0^2 (e^t + e^{-t}) dt - 4\pi \int_0^2 (te^t + te^{-t}) dt \\
 &= 10\pi \left[e^t - e^{-t} \right]_0^2 - 4\pi \left[te^t - e^t - te^{-t} - e^{-t} \right]_0^2 = 10\pi (e^2 - e^{-2}) - 4\pi (2e^2 - e^2 - 2e^{-2} - e^{-2} + 2) \\
 &= 6\pi e^2 + 2\pi e^{-2} - 8\pi .
 \end{aligned}$$